

Answers for the First Midterm Exam

Solution of the Exercise 1.

The mean value theorem has been proven in class.

Solution of the Exercise 2.

We define the following function:

$$N: \mathbb{R}^2 \longrightarrow \mathbb{R}$$

$$(x, y) \longmapsto \sup_{t \in \mathbb{R}} \left| \frac{x + ty}{1 + t^2} \right|.$$

1. First, let $(x, y) \in \mathbb{R}^2$ and $t \in \mathbb{R}$. We have

$$\left| \frac{x + ty}{1 + t^2} \right| \leq \frac{|x|}{1 + t^2} + \left| \frac{t}{1 + t^2} \right| |y|,$$

since $\forall t \in \mathbb{R}, \frac{1}{1 + t^2} \leq 1$ and $\frac{|t|}{1 + t^2} \leq \frac{1}{2}$, then

$$\left| \frac{x + ty}{1 + t^2} \right| \leq |x| + \frac{1}{2}|y| \text{ (which doesn't depend on } t \text{)}.$$

Hence the map $t \mapsto \left| \frac{x + ty}{1 + t^2} \right|$ is bounded and so N is well defined.

Now, let $(x, y), (x', y') \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$. We have

(a) $N(x, y) = 0$ if and only if $(x, y) = (0, 0)$,

(b) $N(\lambda(x, y)) = |\lambda|N(x, y)$,

(c) $N((x, y) + (x', y')) \leq N(x, y) + N(x', y')$.

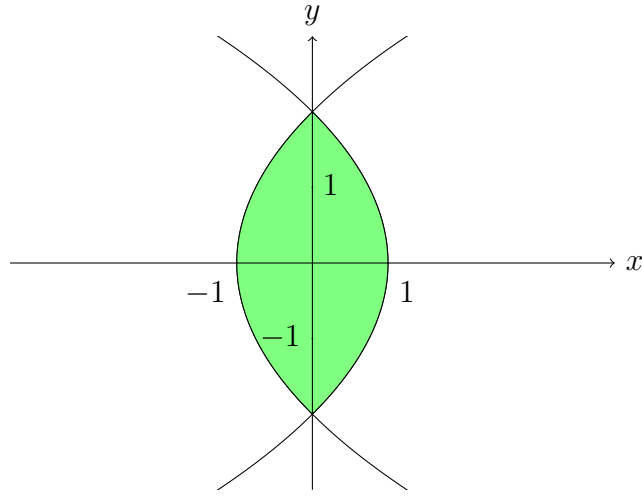
Thus, N defines a norm on $\mathbb{R}^2$.

2. Let $(x, y) \in \mathbb{R}^2$ and $t \in \mathbb{R}$. We have

$$\begin{aligned} \left| \frac{x + ty}{1 + t^2} \right| \leq 1 &\iff |x + ty| \leq 1 + t^2 \\ &\iff -1 - t^2 \leq x + ty \leq 1 + t^2 \\ &\iff t^2 + ty + x + 1 \geq 0 \text{ and } t^2 - ty + 1 - x \geq 0 \\ &\iff \Delta = y^2 - 4(x + 1) \leq 0 \text{ and } \Delta = y^2 - 4(1 - x) \leq 0 \\ &\iff y^2 \leq 4x + 4 \text{ and } y^2 \leq -4x + 4. \end{aligned}$$

Then the unit ball of the norm N is given by:

$$\{(x, y) \in \mathbb{R}^2 \mid (y^2 \leq 4x + 4) \text{ and } (y^2 \leq -4x + 4)\}.$$



3. We put the map $t \mapsto f(t) = \frac{x+ty}{1+t^2}$, where $t \in \mathbb{R}$. By studying the variations of the function f we will have its table of variations as follows:

$y < 0$

t	$-\infty$	β		α	$+\infty$	
$f'(t)$		+	0	-	0	+
f			$f(\beta)$		$f(\alpha)$	0
	0					

$y > 0$

t	$-\infty$	β	α	$+\infty$
$f'(t)$		$-$ 0 $+$ 0 $-$		
f	0 $\searrow$	$f(\beta)$ $\nearrow$	$f(\alpha)$ $\searrow$	0

where $f'(t) = \frac{-yt^2 - 2xt + y}{(1+t^2)^2}$, $t' = \frac{-x + \sqrt{x^2 + y^2}}{y}$, $t'' = \frac{-x - \sqrt{x^2 + y^2}}{y}$, $\alpha = \max\{t', t''\}$ and $\beta = \min\{t', t''\}$. Then, we conclude that:

$$\forall t \in \mathbb{R}, |f(t)| \leq \max\{|f(t')|, |f(t'')|\}.$$

In the other hand, $f(t') = \frac{1}{2} \left(\sqrt{x^2 + y^2} + x \right)$ and $f(t'') = -\frac{1}{2} \left(\sqrt{x^2 + y^2} - x \right)$, hence

$$N(x, y) = \frac{1}{2} \left(\sqrt{x^2 + y^2} + |x| \right).$$

Remark. For the case where $y = 0$, we have $N(x, y) = |x|$.

4. Let $(x, y) \in \mathbb{R}^2$. We have $2N(x, y) = \|(x, y)\|_2 + |x| \geq \|(x, y)\|_2$, then

$$N(x, y) \geq \frac{1}{2} \|(x, y)\|_2.$$

In the other hand, we have $2N(x, y) = \|(x, y)\|_2 + |x| \leq \|(x, y)\|_2 + \|(x, y)\|_\infty \leq \|(x, y)\|_2 + \|(x, y)\|_2 = 2\|(x, y)\|_2$, so

$$N(x, y) \leq \|(x, y)\|_2.$$

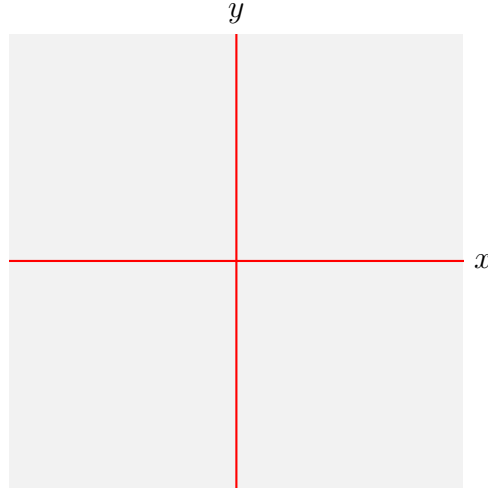
Solution of the Exercise 3.

We define the following function:

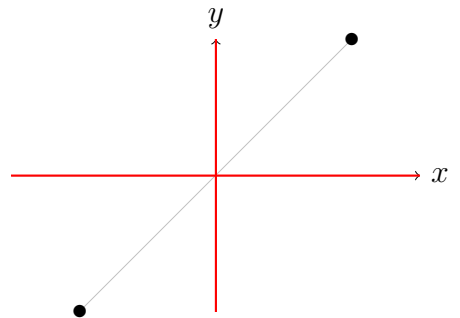
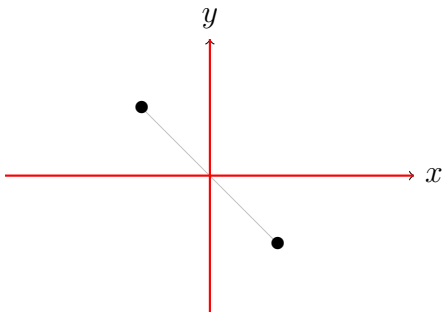
$$f(x, y) = \begin{cases} x^2 \cos\left(\frac{y}{x}\right) + y^2 \cos\left(\frac{x}{y}\right) & \text{if } x \neq 0 \text{ and } y \neq 0 \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}.$$

1. The definition domain of f is given by:

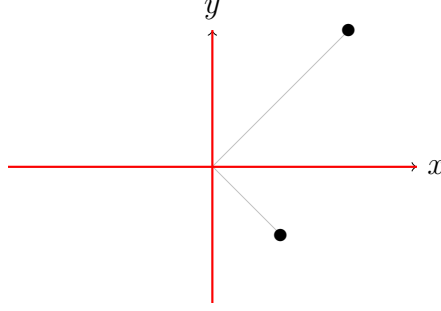
$$\Omega = \{(x, y) \in \mathbb{R}^2 \mid x \neq 0 \text{ and } y \neq 0\} \cup \{(0, 0)\} = \mathbb{R}^* \times \mathbb{R}^* \cup \{(0, 0)\} = \mathbb{R}^2 \setminus \{(x, y) \in \mathbb{R}^2 \mid x = 0 \text{ or } y = 0\}.$$



2. It is clear that Ω is not bounded and then Ω is not compact. Regarding connectedness, it is easy to see that any point in one quarter of the plane can be connected by a segment to another point in the quarter of the plane that is symmetric with respect to the origin.



Furthermore, any point of one quarter of the plane can be connected by two segments (in continuous way) to another point in one quarter of the plane that is symmetric with respect to x -axis or y -axis.



We conclude that our definition domain Ω is connected by arcs and so Ω is connected.

3. In the open subset $\Omega \setminus \{(0,0)\}$, the function f is written as an elementary operation on smooth (C^∞) functions and so f is smooth on $\Omega \setminus \{(0,0)\}$. Therefore f is continuous on $\Omega \setminus \{(0,0)\}$.

For the continuity in $(0,0)$, we have :

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} |f(x,y)| &= \lim_{(x,y) \rightarrow (0,0)} \left| x^2 \cos\left(\frac{y}{x}\right) + y^2 \cos\left(\frac{x}{y}\right) \right| \\ &\leq \lim_{(x,y) \rightarrow (0,0)} x^2 + y^2 \\ &= 0. \end{aligned}$$

Hence $\lim_{(x,y) \rightarrow (0,0)} |f(x,y)| = 0$ and so $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = f(0,0) = 0$. Consequently, f is continuous in $(0,0)$.

4. Let $x_0, y_0 \in \mathbb{R}^*$. We have:

$$\lim_{(x,y) \rightarrow (x_0,0)} f(x,y) = x_0^2 \text{ and } \lim_{(x,y) \rightarrow (0,y_0)} f(x,y) = y_0^2.$$

Thus, f is extendable by continuity on $\mathbb{R}^2$, and we denote its extension $\tilde{f} : \mathbb{R}^2 \rightarrow \mathbb{R}$ is given by:

$$\tilde{f}(x,y) = \begin{cases} x^2 \cos\left(\frac{y}{x}\right) + y^2 \cos\left(\frac{x}{y}\right) & \text{if } x \neq 0 \text{ and } y \neq 0 \\ x^2 & \text{if } x \neq 0 \text{ and } y = 0 \\ y^2 & \text{if } x = 0 \text{ and } y \neq 0 \\ 0 & \text{if } x = 0 \text{ and } y = 0 \end{cases}$$